02-10-2025

Midterm:

↳ 5 questions (± 1)
↳ all material covered fair game
↳ OH Th-Fr

↳ ?'s:
→ problems like HW's (not exact)
→ Proofs
→ best study read notes + book
→ conceptual
↳ diff b/w linear projection + linear CEF
☆ → Interpreting regression

$$\beta = E[XX^T]^{-1} E[XY]$$

$$\hat{\beta} = \left(\frac{1}{n}\sum_{i=1}^{n} x_i x_i^T\right)^{-1} \frac{1}{n}\sum_{i=1}^{n} x_i y_i$$

↳ analogy principle

$$\hat{\beta} = \underset{b}{\operatorname{argmin}} \frac{1}{n}\sum_{i=1}^{n} (y_i - x_i^T b)^2$$

↳ expectations, sums, int's play same role

$$\hat{\beta} = (\underline{X}^T \underline{X})^{-1} (\underline{X}^T \underline{Y})$$

↳ model.matrix in R

$$\underset{k\times n}{\underline{X}^T}\,\underset{n\times k}{\underline{X}} = (x_1, x_2, \ldots, x_n)\begin{pmatrix} x_1^T \\ x_2^T \\ \vdots \\ x_n^T \end{pmatrix} = \sum_{i=1}^{n} x_i^T x_i$$

$$\underline{X}^T \underline{Y} = (x_1, \ldots, x_n)\begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} = \sum_{i=1}^{n} x_i^T y_i$$

Fitted Value:

↳ $\hat{Y}_i = x_i^T \hat{\beta}$

Residual:

↳ $\hat{e}_i = y_i - \hat{y}_i = y_i - x_i^T \hat{\beta}$

↳ properties:

→ $\frac{1}{n}\sum_{i=1}^{n} x_i \hat{e}_i = 0$

→ $\frac{1}{n}\sum_{i=1}^{n} \hat{e}_i = 0$ if x_i includes an intercept

→ $y_i = x_i^T \hat{\beta} + \hat{e}_i$

→ $\underline{\hat{e}} = \begin{pmatrix} \hat{e}_1 \\ \vdots \\ \hat{e}_n \end{pmatrix} = \underline{y} - \underline{X}\hat{\beta}$

Projection Matrices:

↳ def: $\underline{P} = \underline{X} \, (\underbrace{\underline{X}^T \underline{X}}_{k \times k})^{-1} \underbrace{\underline{X}^T}_{k \times n}$. P creates fitted values.

$n \times k$

s.t. $\underline{P}\,\underline{Y} = \underline{X} (\underline{X}^T \underline{X})^{-1} \underline{X}^T \underline{Y} = \underline{X} \hat{\beta}_{n \times 1}$

↳ properties

1. $\underline{P}\underline{X} = \underline{X}$
2. $\underline{P}$ is symmetric $\Rightarrow \underline{P}^T = \underline{P}$
3. $\underline{P}$ is idempotent $\Rightarrow (\underline{P}\,\underline{P} = \underline{P})$
4. $tr(\underline{P}) = k$
5. $\underline{P}$ is P.S.D

↳ proof 1: $\underline{P}\,\underline{X} = \underline{X} \underbrace{(\underline{X}^T \underline{X})^{-1} (\underline{X}^T \underline{X})}_{\text{cancel}} = \underline{X}$

↳ proof 2: $\underline{P}^T = (\underline{X} (\underline{X}^T \underline{X})^{-1} \underline{X}^T)^T = (\underline{X}^T)^T (((\underline{X}^T \underline{X})^{-1})^T (\underline{X})^T$

$= \underline{X} \, ((\underline{X}^T \underline{X})^T)^{-1} \underline{X}^T = \underline{X} ((\underline{X})^T (\underline{X}^T)^T)^{-1} \underline{X}^T$

$= \underline{X} \, (\underline{X}^T \underline{X})^{-1} \underline{X}^T = \underline{P}$

↳ can swap middle term for pos. def & symmetric

↳ proof 3: $\underline{P} \, \underline{X} (\underline{X}^T \underline{X})^{-1} \underline{X}^T$. By property 1 $= X = \underline{P}$

↳ proof 4: $tr(\underline{P}) = tr(\underbrace{\underline{X} (\underline{X}^T \underline{X})^{-1}}_{A_{n \times k}} \underbrace{\underline{X}^T}_{B_{k \times n}}) = tr(\underline{X}^T \underline{X} (\underline{X}^T \underline{X})^{-1})$

$= tr(\underline{I}_k) = k$

↳ proof 5: $c_{n \times 1}$: $c^T \underline{P} c \geq 0 \Rightarrow c^T \underline{P} c = c^T \underline{P}\,\underline{P} c =$ 3

$c^T \underline{P}^T \underline{P} c = (\underline{P} c)^T \underline{P} c \geq 0$

2

Annihilator Matrix:

$\hookrightarrow \underset{n\times n}{\underline{M}} := \underline{I}_n - \underline{P} = \underline{I} - X(X^T X)^{-1} X^T$

$\hookrightarrow$ Properties:

① $\underline{M}\,\underline{X} = (\underline{I} - \underline{P})\underline{X} = \underline{X} - \underline{X} = 0$

② $\underline{MY} = (\underline{I} - \underline{P})\underline{Y} = \underline{Y} - \underline{P}\,\underline{Y} = \underline{Y} - \underline{X}\hat{\beta} = \hat{e}$

$\underline{\hat{e}} = \underline{M}\,\underline{Y} = \underline{M}(\underline{X}\beta + \underline{e}) = 0 + \underline{M}\,\underline{e} = \underline{M}\,\underline{e}$

Estimation of Error Variance:

$\hookrightarrow \sigma^2 := \operatorname{var}(e) = E[e^2]$

$\hookrightarrow \tilde{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n} e_i^2$ (infeasible)

$\hookrightarrow \hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n} \hat{e}_i^2$ (feasible)

Regression Components:

$\underset{n\times k}{\underline{X}} = [\underset{n\times k_1}{\underline{X}_1} \;\; \underset{n\times k_2}{\underline{X}_2}]; \quad \beta = \begin{pmatrix} \beta_1 \\ \beta_2 \end{pmatrix}$

$\underline{Y} := [\underline{X}_1 \; \underline{X}_2]\begin{pmatrix} \hat{\beta}_1 \\ \hat{\beta}_2 \end{pmatrix} + \hat{e} = \underline{X}_1\hat{\beta}_1 + \underline{X}_2\hat{\beta}_2 + \hat{e}$

$$\hat{\beta} = \begin{pmatrix} \hat{\beta}_1 \\ \hat{\beta}_2 \end{pmatrix} = \underset{b_1,\, b_2}{\operatorname{argmin}} \sum_{i=1}^{n} \left(y_i - x_{i1}^T b_1 - x_{i2}^T b_2\right)^2$$

$$= \underset{b_1,\, b_2}{\operatorname{argmin}} (\underline{y} - \underline{X}_1 b_1 - \underline{X}_2 b_2)^T (\underline{y} - \underline{X}_1 b_1 - \underline{X}_2 b_2)$$

$$= \underset{b_1}{\operatorname{argmin}} \left\{ \overbrace{\min_{b_2} (y - \underline{X}_1 b_1 - \underline{X}_2 b_2)^T (\underline{y} - \underline{X}_1 b_1 - \underline{X}_2 b_2)}^{*} \right\}$$

$$\hat{\beta}_2(b_1) = (\underline{X}_2^T \underline{X}_2)^{-1} \underline{X}_2^T (\underline{y} - \underline{X}_1 b_1)$$

$$\star = \underbrace{(\underline{y} - \underline{X}_1 b_1 - \underline{X}_2 \hat{\beta}_2(b_1))}_{\star\star}{}^T (\underline{y} - \underline{X}_1 b_1 - \underline{X}_2 \hat{\beta}_2(b_1))$$

$$\star\star = y - \underline{X}_1 b_1 - \underbrace{X_2 (X_2^T X_2)^{-1} X_2^T}_{\underline{P}_2} (\underline{y} - \underline{X}_1 b_1) = \underbrace{I - \underline{P}_2}_{m_2} (y - \underline{X}_1 b_1)$$

$$= \underline{m}_2 (y - X_1 b_1)$$

$$\star = (\underline{m}_2 (\underline{y} - \underline{X}_1 b_1))^T (\underline{m}_2 (\underline{y} - \underline{X}_1 b_1))$$

$$= (\underline{y} - \underline{X}_1 b_1)^T \underline{m}_2^T \underline{m}_2 (\underline{y} - \underline{X}_1 b_1) = (\underline{y} - \underline{X}_1 b_1)^T \underline{m}_2 (\underline{y} - \underline{X}_1 b_1)$$

$$\underset{b_1}{\operatorname{argmin}} (\underline{y} - \underline{X}_1 b_1)^T \underline{m}_2 (\underline{y} - \underline{X}_1 b_1) = \underline{y}^T \underline{m}_2 \underline{y} - 2 b_1^T \underline{X}_1^T \underline{m}_2 \underline{y} + b_1^T \underline{X}_1^T \underline{m}_2 \underline{X}_1 b_1$$

F.O.C.

$$0 = -2 \underline{X}_1^T \underline{m}_2 \underline{y} + 2 \underline{X}_1^T \underline{m}_2 \underline{X}_1 \hat{\beta}_1$$

$$\hat{\beta}_1 = (\underline{X}_1^T \underline{m}_2 \underline{X}_1)^{-1} \underline{X}_1^T \underline{m}_2 \underline{y}$$

$$= (\underline{X}_1^T \underline{m}_2^T \underline{m}_2 \underline{X}_1)^{-1} \underline{X}_1^T \underline{m}_2^T \underline{m}_2 \underline{y}$$

$$\star = (\tilde{\underline{X}}_1^T \tilde{\underline{X}})^{-1} \tilde{\underline{X}}^T \tilde{e}_2 \quad \text{s.t.} \quad \begin{aligned} \tilde{\underline{X}}_1 &= \underline{m}_2 X_1 \\ \tilde{e}_2 &= \underline{m}_2 \underline{y} \end{aligned}$$

Frisch Waugh Lovell Thm